High-Level Design – Assistant

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** High-Level Design (Sub-module: TFA Assistant, code `TFA`) · **Version:** 1.0 · **Status:** Draft
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> **Based on:** BRD-Assistant v1.0, FRD-Assistant v1.0
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Document Control

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version

Related documents:

Document	Location
Overall HLD (stack, deployment, cross-cutting design, decisions `D-nn`)	`../../00-Overall/HLD-TaskFlow-Overall.md`
BRD – Assistant	`../../../BRD/08-TFA-Assistant/08.1-Assistant/BRD-Assistant.md`
FRD – Assistant (`FR-TFA-nn`, `UC-TFA-nn`)	`../../../FRD/08-TFA-Assistant/08.1-Assistant/FRD-Assistant.md`
LLD – Assistant (classes, endpoints, tables)	`../../../LLD/08-TFA-Assistant/08.1-Assistant/LLD-Assistant.md`
HLD – TFA Settings (provider, model and API key configuration)	`../../06-Settings/06.3-TFA-Settings/HLD-TFA-Settings.md`
HLD – Mail, My Profile, Employee Management (targets of confirmed actions)	`../../04-Communication/04.1-Mail/HLD-Mail.md`, `../../07-My-Profile/07.1-My-Profile/HLD-My-Profile.md`, `../../03-TaskFlow-Management/03.1-Employee-Management/HLD-Employee-Management.md`
TDD v2.0 – §16 TFA Assistant Design	`TF - Documents/TF - TDD/TDD-TaskFlow.md`

1. Introduction

1.1 Purpose

This HLD describes how TFA (TaskFlow Assistant) is built: a tool-calling chat assistant hosted inside the TaskFlow API, with a panel on every signed-in screen. It covers the conversation engine, the model gateway to Ollama, Gemini and OpenAI, the permission-gated tools, the confirm-before-act mechanism for actions, the built-in help knowledge, voice input in the browser, and the safety controls (secret screen, rate limit, audit). Classes, endpoint details and columns are in the LLD.

1.2 Scope

In scope: TFA Panel (conversation, model selector, cloud notice, new chat, provider failures), task answers and the ranked work order, how-to help from 19 topics, Action Cards (mail draft, own-profile update, role change), user lookup and fixed guidance, voice controls, message limit and action audit.

Out of scope: configuring providers, models and API keys (TFA Settings HLD); the mailbox, profile and employee rules themselves (their HLDs); creating, editing or reassigning tasks through TFA; password changes through TFA; an audit screen. Shared stack and deployment are in the overall HLD.

Change from original BRD: the original BRD has no assistant; TFA is an addition (TDD D11). It uses the BRD's existing rules through the target modules (Mail, My Profile, Employee Management, Task Management).

1.3 Audience

Developer, tester and reviewer of TaskFlow; the product owner (Jojo George).

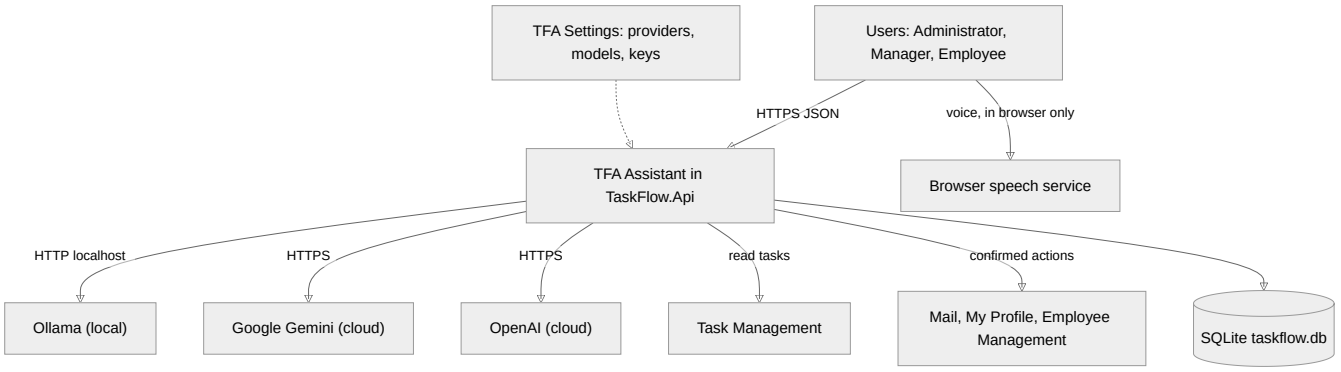
1.4 Definitions & Abbreviations

Term	Meaning
Provider	AI service that writes answers: Ollama (local), Gemini or OpenAI (cloud) – `AiProviderKind`
Tool	Server function the model may ask to run (read tool or action tool)
Tool round	One model call that may request tools; at most 4 per answer
Pending action	Prepared action stored with its exact payload, waiting for the user's confirm (10-minute lifetime)
Action Card	Panel card that shows a pending action with Confirm / Edit / Cancel
Secret screen	Check that stops passwords and API keys from reaching a provider or the database
SPA	Single Page Application (Angular frontend)

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2. System Context

2.1 Context Diagram



2.2 Actors & External Systems

Actor / system	Interaction	Protocol
Users with `assistant.chat`	Chat, read answers, ranked tasks, help	HTTPS / REST (JSON)
Users with `assistant.actions.confirm` (and `roles.manage` for role changes)	Edit, confirm or cancel Action Cards	HTTPS / REST
Ollama	Local chat model (default `llama3.2:3b`)	HTTP to `{BaseUrl or http://localhost:11434}/api/chat`
Google Gemini	Cloud chat model (default `gemini-2.0-flash`)	HTTPS, key in `x-goog-api-key` header
OpenAI	Cloud chat model (default `gpt-4o-mini`)	HTTPS, Bearer key
Browser speech service	Wake phrase, dictation, read-aloud (Chrome, Edge)	Web Speech API in the browser; nothing sent to TaskFlow but recognised text
TFA Settings	Which providers are enabled, model names, encrypted keys	In-process reads
Task Management, Mail, My Profile, Employee Management	Data for answers; execution of confirmed actions	In-process service calls

3. Architecture Goals, Constraints & Principles

3.1 Quality Goals

Goal	How the architecture supports it
Never exceed the user's rights	Tools advertised and executed only when the caller passes the same permission policies as the screens; re-checked at confirm (NFR-TFA-02)

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Goal	How the architecture supports it
Human in control	Action tools only prepare; only the user's confirm call executes (D-TFA-2)
Trustworthy answers	Ranking computed in code; how-to only from built-in topics (D-TFA-4, D-TFA-5)
Privacy	Secret screen before any provider call; local provider option; cloud notice (NFR-TFA-01, NFR-TFA-03)
Resilience to provider problems	Classified failures, plain messages, Retry / other provider, nothing stored on failure (NFR-TFA-04, NFR-TFA-08)
Fair use	20 messages per user per minute (NFR-TFA-06)

3.2 Constraints

- Single server; rate limiter and provider list refresh are per process.
- Providers are external and may be slow, rate-limited or offline; HTTP timeout `AiProviders:TimeoutSeconds` (60).
- Voice depends on the browser's speech recognition (Chrome, Edge).
- The model is untrusted: its output and the data it reads must never be able to change permissions or run actions.

3.3 Architecture Principles

- **The model proposes, the system disposes:** permissions, ranking, help selection and action execution are code, not model decisions.
- **Caller from the token only:** no tool accepts a user id, role or password.
- **Data is data:** task, comment and mail text returned by tools is marked as data and redacted for secrets before reaching the model.
- **Reuse target rules:** a confirmed action runs through the target module's own validator and service.

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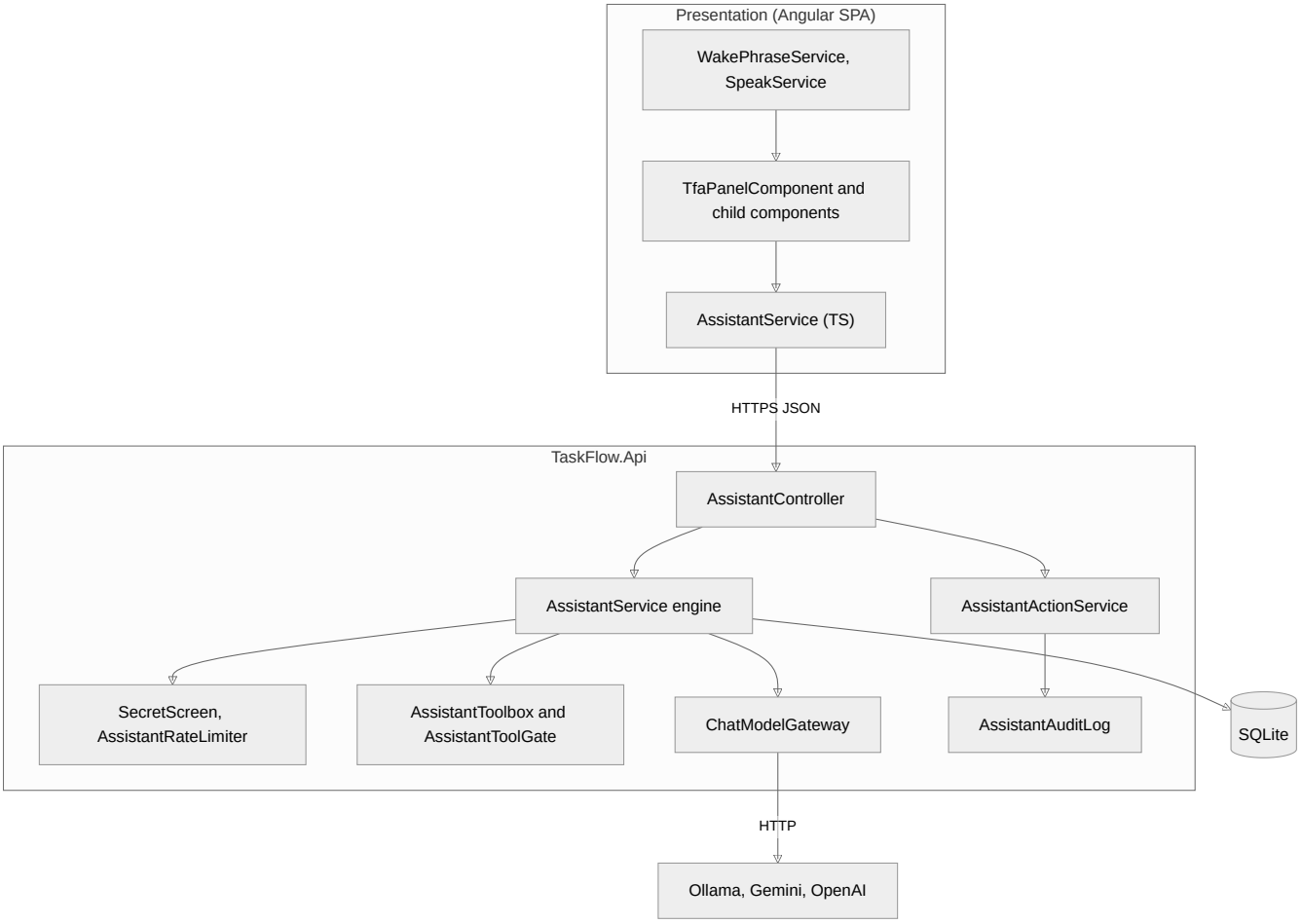
4. Architecture Overview

4.1 Architecture Style

Assistant feature inside the TaskFlow three-tier application: Angular panel components → REST controller under `/api/v1/assistant` → conversation engine with a bounded tool loop → model gateway with pluggable provider adapters (typed `HttpClient`s) → tools that call other modules' services; pending actions and audit stored in SQLite. The shared pipeline is in the overall HLD.

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4.2 Logical Architecture



4.3 Layers & Responsibilities

Layer	Responsibility	May call
Presentation	`TfaPanelComponent` (button, panel, feed, cloud notice, failure card), `TfaHeaderComponent` (new chat, close, model selector), `TfaComposerComponent` (message box, suggestions, mic), `TfaMessageComponent` (text, ranked list, links, cards), `TfaActionCardComponent` (confirm / edit / cancel); `AssistantService` (TS) signals; `WakePhraseService`, `SpeakService`; browser storage `tfa.provider`, `tfa.cloudAck`, `tfa.handsfree`	API, browser speech
API / Controllers	`AssistantController`: caller from the token (`AssistantCaller`), `perm:assistant.chat` / `perm:assistant.actions.confirm` per action	Services
Conversation engine	`AssistantService`, `AssistantPrompt`, `AssistantTexts`, `ProviderFailureMessages`: length check, secret screen, provider choice, rate limit, history window, tool loop, persistence	Gateway, tools, repositories
Tools	`AssistantToolbox` + `AssistantToolGate` (via `IAssistantAuthorizer`); read tools (tasks, ranking, help, user lookup, guidance) and action tools (draft mail, profile update, role change)	Other modules' services, `HelpKnowledgeBase`, `TaskPrioritiser`

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Layer	Responsibility	May call
Actions	`AssistantActionService`, `AssistantActionRules`: ownership, expiry, permission re-check, single claim, execution, audit	Mail, Profile, Employee services
Model gateway / integrations	`ChatModelGateway` → `ChatModelProviderRegistry` → `OllamaProvider`, `GeminiProvider`, `OpenAiProvider`; `ProviderHttp`, `ProviderFailure`, `AiProviderRuntime`	External providers
Repositories	Conversation, message, pending action, audit, directory repositories	Database

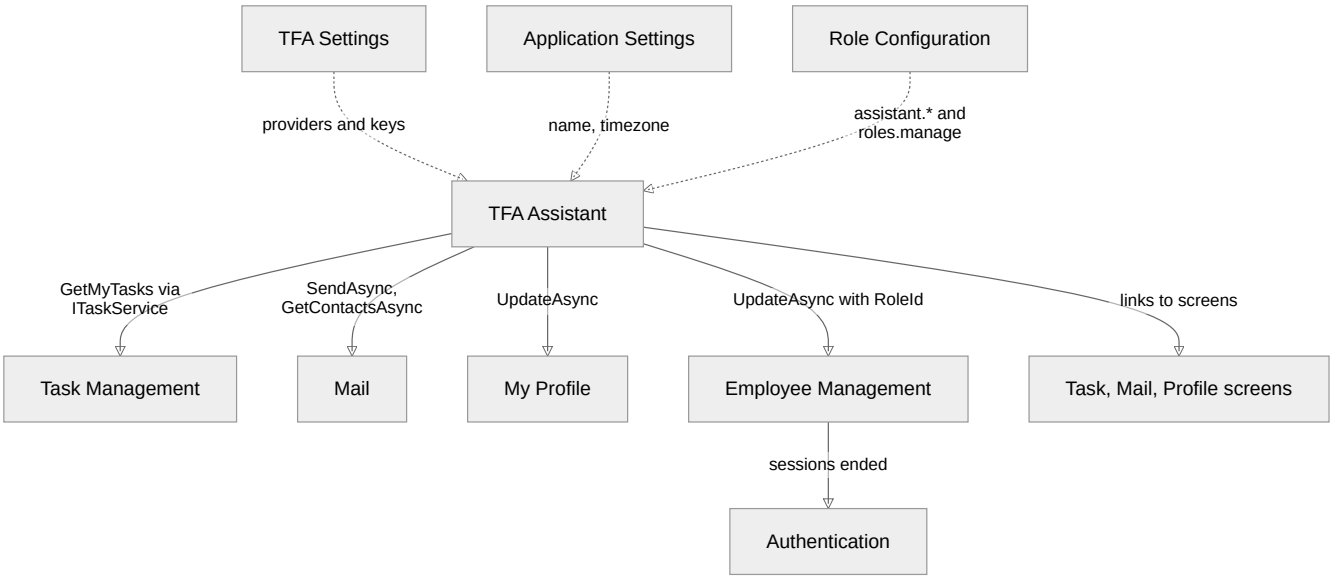
5. Component / Module Design

Module	Responsibility	Key components (high level)	Depends on
TFA Panel (chat)	Panel on every signed-in screen; conversation load, send, new chat, model selector, cloud notice, provider failure card	`TfaPanelComponent`, `TfaHeaderComponent`, `TfaComposerComponent`, `TfaMessageComponent`, `AssistantService` (TS); `AssistantController`; `AssistantService` engine	TFA Settings (configured providers)
Conversation engine	Validate, screen secrets, limit rate, build prompt and history, run up to 4 tool rounds, store question and answer	`AssistantService`, `AssistantPrompt`, `SecretScreen`, `AssistantRateLimiter`, conversation and message repositories	Application Settings (name, timezone)
Model gateway	One interface over three providers; failure classification (`ProviderFailureKind`)	`ChatModelGateway`, `ChatModelProviderRegistry`, provider adapters	TFA Settings (`AiProviderConfig`, `AiModel`, encrypted keys via `AiKeyProtector`)
Task answers and work order	Own / accessible open tasks; deterministic ranking	`GetMyTasksTool`, `PrioritiseTasksTool`, `TaskPrioritiser`	Task Management (`ITaskService`, access rules)
How-to help	Keyword-scored, role-aware answers from 19 embedded topics with screen links	`SearchHelpTool`, `HelpKnowledgeBase`	–
Lookup and guidance	Find users for mail; fixed password and "needs an administrator" guidance	`FindUserTool`, `ExplainPasswordChangeTool`, `ContactAdminGuidanceTool`	Mail (`IMailService` contacts)
Action Cards	Prepare, edit (mail only), confirm, cancel; expiry; run once; audit	`DraftMailTool`, `ProposeProfileUpdateTool`, `ProposeRoleChangeTool`, `AssistantActionService`, `TfaActionCardComponent`, pending-action repository	Mail, My Profile, Employee Management services and validators
Voice controls	Wake phrase, dictation, read-aloud, hands-free toggle	`WakePhraseService`, `SpeakService` (frontend only)	Browser Web Speech API

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Module	Responsibility	Key components (high level)	Depends on
Background safeguards	Rate limit, outcome audit	`AssistantRateLimiter`, `AssistantAuditLog`, audit repository	–

5.1 Module Interaction



From → To	What	Coupling
TFA → Task Management	Open tasks in the user's scope (`mine` or `accessible`, max 20)	`ITaskService`, so Task Management scope applies (BR-TFA-05)
TFA → Mail	Contacts lookup; send a confirmed draft from the user's mailbox	`IMailService` + `SendMailRequestValidator` (BR-TFA-09)
TFA → My Profile	Apply a confirmed own name / email change, mobile unchanged	`IProfileService.UpdateAsync` + `UpdateOwnProfileRequestValidator`
TFA → Employee Management	Apply a confirmed role change; target user's sessions end	`IEmployeeService.UpdateAsync`
TFA Settings → TFA	Enabled providers, model names, keys	Read per request; panel refreshes the list every 60 s

6. Data Architecture (high level)

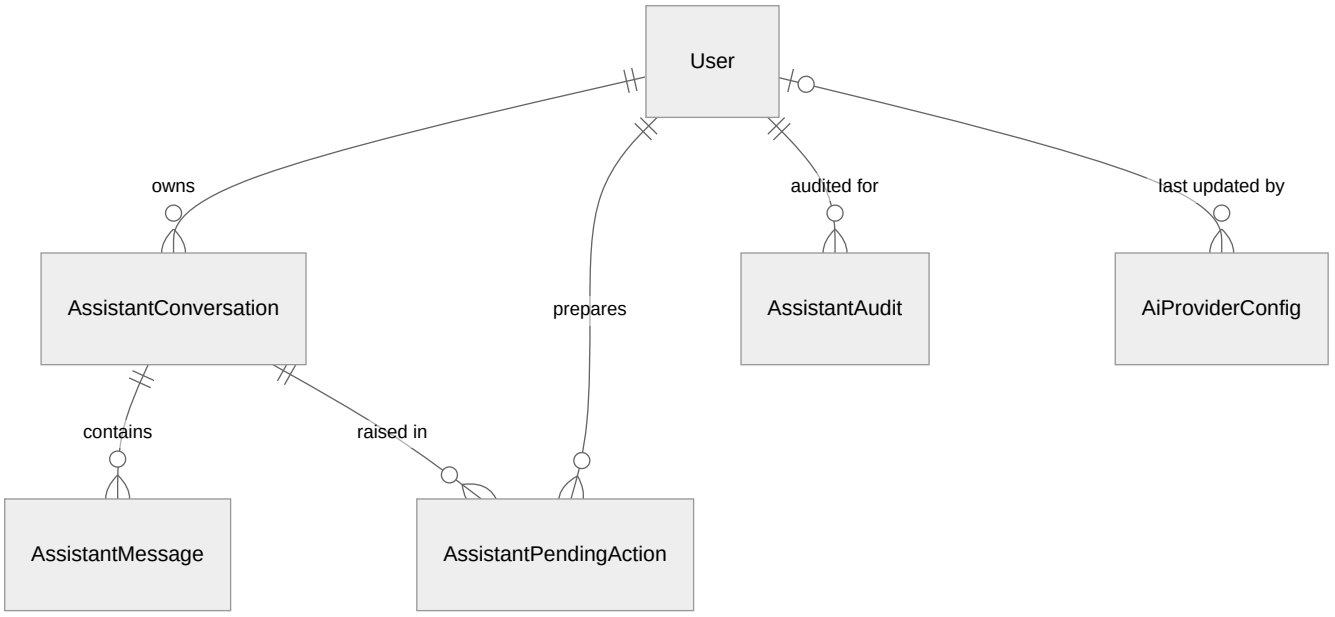
6.1 Data Stores

Store	Technology	Holds	Notes
Main database	SQLite `taskflow.db` (EF Core)	Conversations, messages, pending actions, audit records; reads provider configuration	Migrations `20261018183210_AddAssistant`, `20261019163241_AddAiModels`; conversations kept indefinitely

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Store	Technology	Holds	Notes
Embedded resources	Assembly `TaskFlow.Api`	19 help topics (`Services/Assistant/Knowledge/*.md`)	Read only, shipped with the application
In-memory	Process	Rate-limit windows	Reset on restart
Browser storage	`localStorage`	Chosen provider, cloud acknowledgements, hands-free off	Not on the server

6.2 Main Data Domains



Domain	Owned / used	Purpose
`AssistantConversation`	Owned	A user's chat thread; the latest is active
`AssistantMessage`	Owned	Question, answer or tool message (`AssistantMessageRole`: User, Assistant, Tool) with provider used and display metadata
`AssistantPendingAction`	Owned	Prepared action (`AssistantActionKind`: SendMail, UpdateOwnProfile, ChangeUserRole) with exact payload, `AssistantActionStatus` (Pending, Confirmed, Cancelled, Expired, Refused, Failed) and expiry
`AssistantAudit`	Owned	Outcome record: user, kind, outcome, time – no bodies
`AiProviderConfig`, `AiModel`	Used (TFA Settings)	Enabled providers (`AiProviderKind`: Ollama, Gemini, OpenAI), model names, encrypted keys
`User`, `Task`, `MailMessage`, `Role`	Used through other modules' services	Answers and action targets

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6.3 Data Flow

- **Question:** panel → length and secret checks → rate limit → last 12 messages (about 6000 characters) + prompt → provider, with tool rounds whose results are redacted → question and answer stored together only after a successful answer; a provider failure stores nothing.
- **Action:** action tool stores a pending action with the payload shown on the card → user confirms → ownership, expiry and permission re-checked → single atomic claim → target service executes → status and audit outcome stored.
- **Voice:** browser recognises speech locally (browser service) → text goes through the normal send path; no audio reaches TaskFlow.

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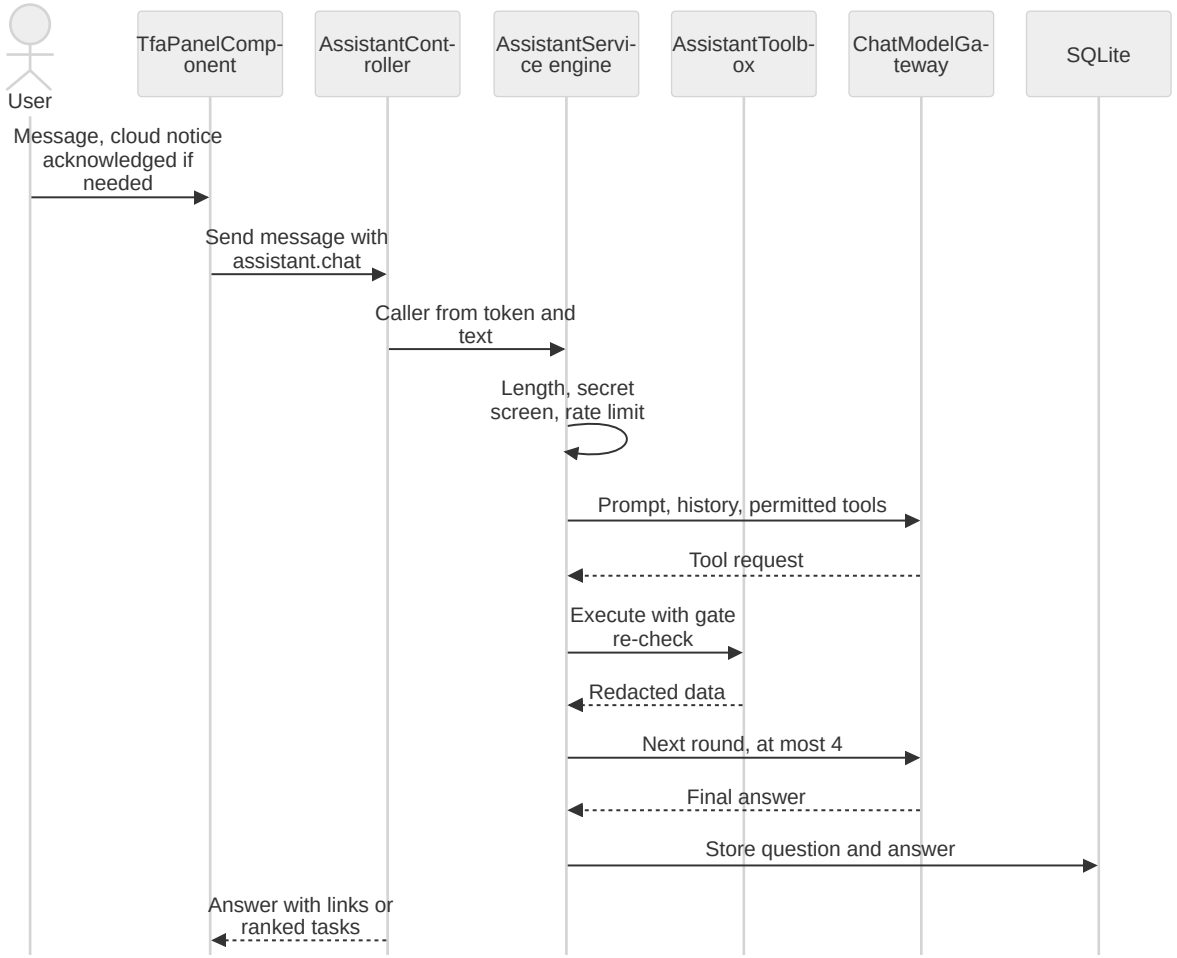
7. Integration Design

Integration	Direction	Purpose	Failure behaviour
Ollama (`OllamaProvider`)	Outbound HTTP	Local answers	Unreachable / model missing → plain message, Retry, Use another provider; nothing stored
Gemini (`GeminiProvider`)	Outbound HTTPS	Cloud answers	Rate limited, invalid key, timeout, unreachable, model unavailable, unexpected → `ProviderFailureKind` message; no automatic switch
OpenAI (`OpenAiProvider`)	Outbound HTTPS	Cloud answers	As Gemini
All providers	–	Typed `HttpClient`s, timeout 60 s, redirects not followed, no header or body logging	Raw provider bodies never shown or logged (provider, kind, HTTP status only)
Browser speech (`WakePhraseService`, `SpeakService`)	Browser local	Wake phrase, dictation, read-aloud	Matching message (unsupported browser, microphone blocked or missing, no internet); typing still works
Mail / My Profile / Employee Management	Outbound in-process	Execute confirmed actions	Status `Failed`, "I could not do that: ... Nothing was sent." / "Nothing was changed."
Task Management	Outbound in-process	Task answers	Inaccessible task → "That task was not found, or you do not have access to it."

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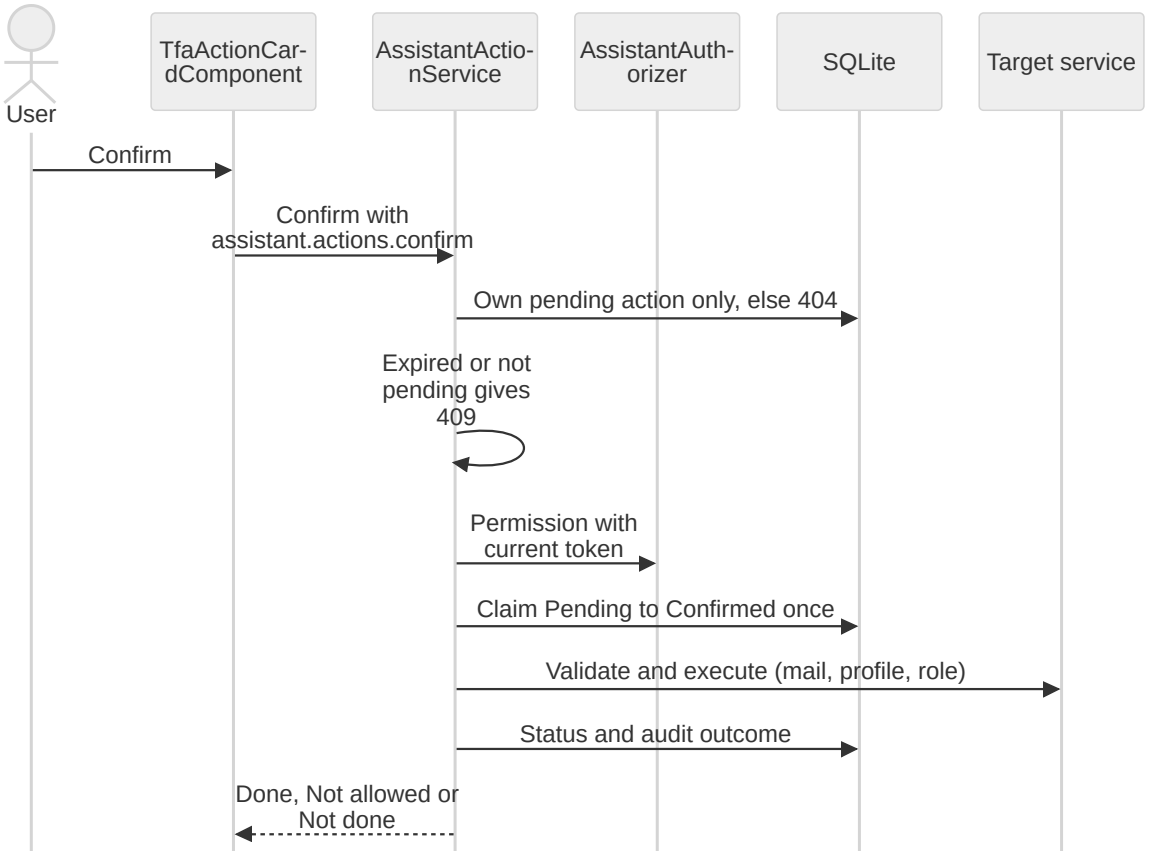
8. Key Flows (high level)

8.1 Ask a question (UC-TFA-01, main business flow)



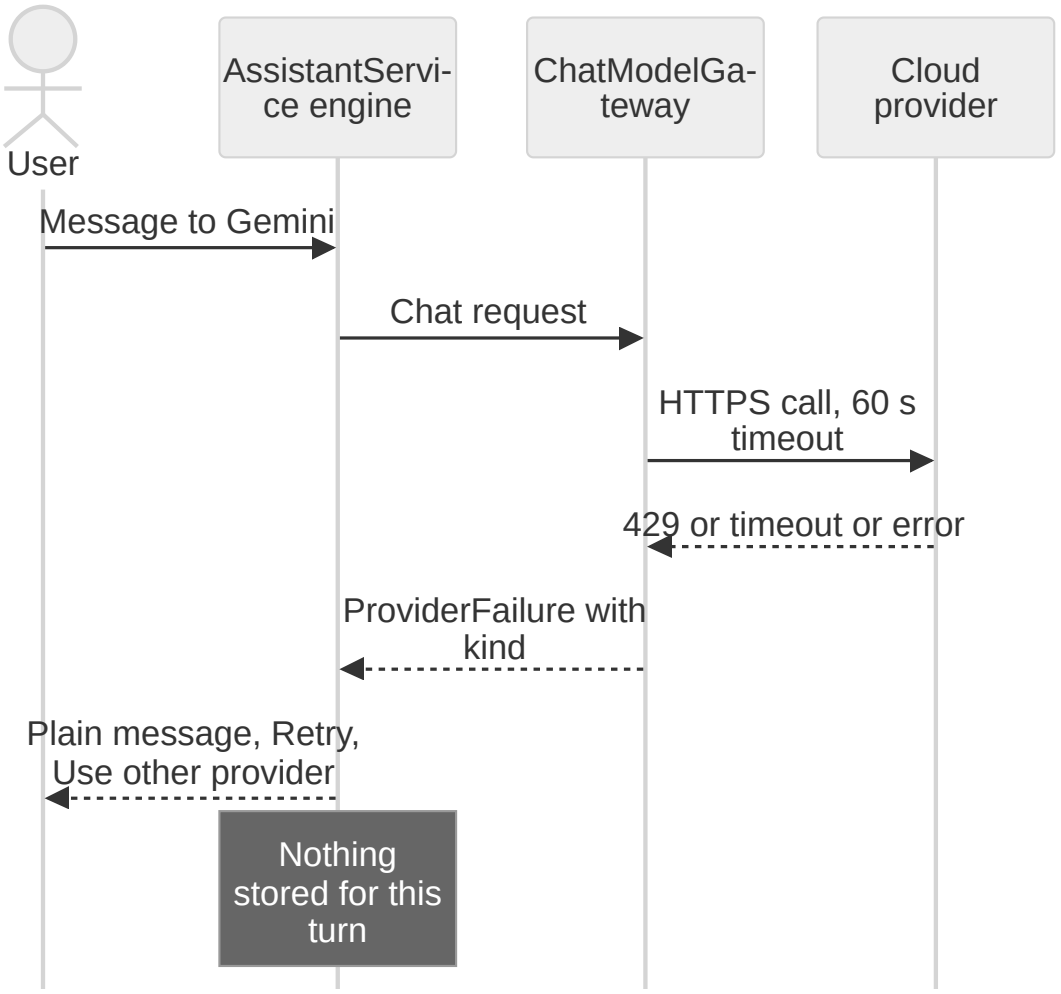
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8.2 Confirm a prepared action (UC-TFA-05 – UC-TFA-07)

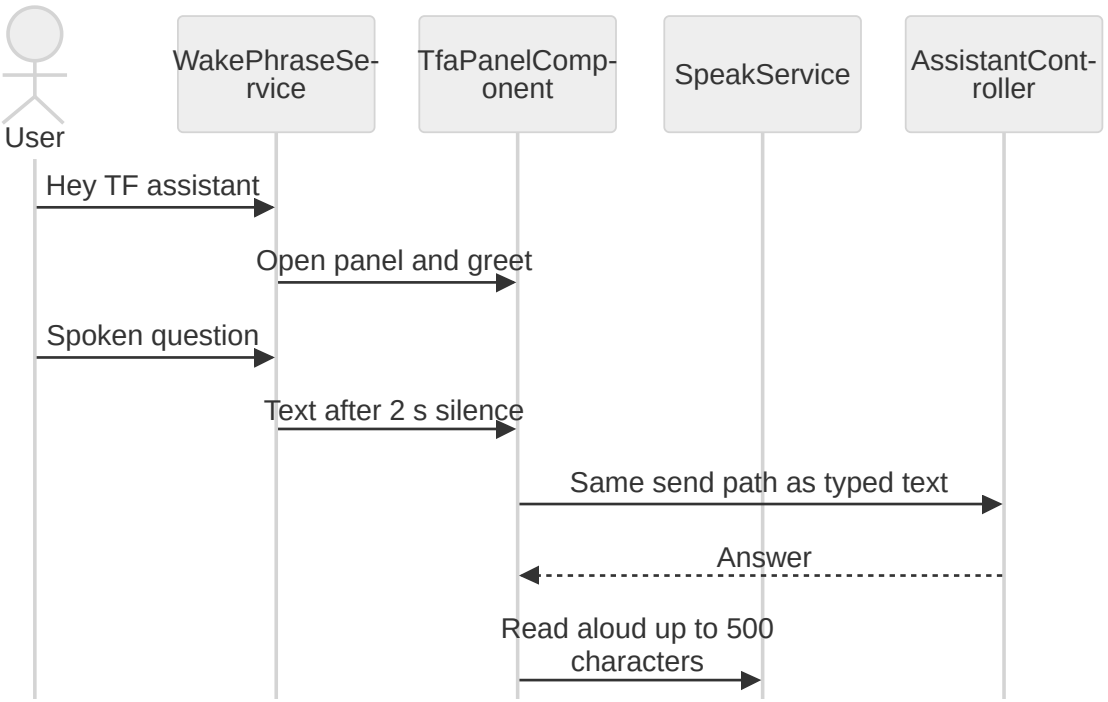


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8.3 Provider failure (integration)



8.4 Ask by voice (UC-TFA-08)



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9. Security Architecture

Concern	Approach
Authentication	Shared JWT bearer + refresh cookie, session validation, MFA setup gate (overall HLD)
Authorization	`perm:assistant.chat` on conversation, messages, new chat and providers; `perm:assistant.actions.confirm` on edit, confirm and cancel; `roles.manage` required by the role-change tool and its confirm. `AssistantAuthorizer` evaluates the API's own `perm:{key}` policies through `IAuthorizationService`, so TFA can never do more than the screens. The panel is mounted by `ShellComponent` only with `assistant.chat`
Data scope	Own conversation and actions only; foreign ids (Guids) return 404, Administrators included; task answers follow Task Management access
Model containment	The model sees only tools the caller may use; every tool call re-checks the gate; action tools only prepare; refusals and confirmations use fixed texts; tool output treated as data (prompt-injection defence, R-TFA-02)
Secrets	`SecretScreen` blocks messages that look like passwords, API keys, Bearer tokens or JWTs before any provider call or storage; tool output passed through redaction; provider keys stored encrypted by TFA Settings (`AiKeyProtector`) and never sent to the SPA
Cloud data	One-time cloud notice per provider; local Ollama option where nothing leaves the machine (R-TFA-03)
Abuse	`AssistantRateLimiter`: 20 messages per user per sliding minute → 429 with `Retry-After`
Auditing	`AssistantAuditLog` records prepared, confirmed, cancelled, expired, refused and failed outcomes without bodies; 403s also logged by `DeniedActionAuditHandler`

10. Technology Stack

TFA uses the shared TaskFlow stack: ASP.NET Core on .NET 10 (`net10.0`) with typed `HttpClient`s for the three providers (no vendor AI SDK is installed), EF Core 10.0.12 with SQLite, FluentValidation 12.2.0 (reused target validators), ASP.NET Core Data Protection for provider keys (TFA Settings), Angular 19.2 with signals for the panel, and the browser Web Speech API for voice (no package). Versions and rationale: overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md`).

11. Deployment Architecture

TFA runs inside the single `TaskFlow.Api` process. Deployment-specific items: an optional Ollama service on the same machine or network (`BaseUrl`, default `http://localhost:11434`), outbound internet access only when Gemini or OpenAI are enabled, `AiProviders:TimeoutSeconds` (60) and the Data Protection key ring (`AiProviders:KeyRingPath`) that holds the keys protecting provider API keys. On a new installation `AiProviderSeeder` enables Ollama with `llama3.2:3b`; Gemini and OpenAI start off without a key. Environments and the deployment diagram: overall HLD (`../../00-Overall/HLD-TaskFlow-Overall.md`).

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12. Non-Functional Design

Area	Design decision
Performance	History window of 12 messages / about 6000 characters; at most 4 tool rounds; conversation load limited to the last 100 messages; task tool max 20, ranking max 10
Scalability	Single instance; provider throughput is the limit; rate limiter protects quotas (NFR-TFA-06)
Reliability	Atomic claim prevents double execution (NFR-TFA-05); nothing stored on provider failure, Retry resends the same text (NFR-TFA-04)
Availability	Local Ollama works offline; cloud providers optional; plain message and next step for every failure (NFR-TFA-08)
Security	Secret screen on every message (NFR-TFA-01); permission re-check at confirm (NFR-TFA-02)
Maintainability	Provider adapters behind one gateway; help topics are markdown files with front matter; tools registered in one toolbox
Observability	Audit outcomes; provider failures logged with provider, kind and HTTP status only
Compatibility	Voice in Chrome and Edge; typing in every supported browser (NFR-TFA-07)
Backup & Recovery	Conversations, actions and audit in `taskflow.db`; key ring folder backed up with it

13. Design Decisions

ID	Decision	Alternatives considered	Rationale
D-TFA-1	Tool-calling assistant hosted in the API; the model sees only tools the caller may use and every call is re-checked	Model with direct database access; free-form answers only	Answers grounded in TaskFlow data; the model can never exceed the user's rights (BR-TFA-01)
D-TFA-2	Confirm-before-act: action tools store a pending action with the exact payload; only the user's confirm executes it; 10-minute expiry; single atomic claim	Let the model execute actions; client-side confirm only	Human control, no replay, no double execution (BR-TFA-02, BR-TFA-08)
D-TFA-3	Permission checks evaluate the API's own `perm:{key}` policies with the caller's current token, at prepare and at confirm	Separate assistant permission logic	One source of truth; a permission removed after preparing makes the confirm fail
D-TFA-4	Work order calculated by `TaskPrioritiser`; the model only explains it	Ask the model to rank	Same order for every provider (BR-TFA-06)
D-TFA-5	How-to answers only from 19 embedded, role-aware markdown topics	Let the model answer from its own knowledge	No invented screens or features (BR-TFA-07)
D-TFA-6	Secret screen before provider and storage; redaction of tool output	Rely on user care	Passwords and keys never leave the browser request (BR-TFA-03)

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ID	Decision	Alternatives considered	Rationale
D-TFA-7	Pluggable providers (Ollama, Gemini, OpenAI) behind `ChatModelGateway` with classified failures and no automatic fallback	Single vendor; silent fallback to another provider	Free local option; the user decides where data goes (BR-TFA-11)
D-TFA-8	Per-user in-memory rate limit of 20 messages per minute	Distributed limiter; no limit	Protects provider quotas on a single server (BR-TFA-12)
D-TFA-9	Voice entirely in the browser (Web Speech API); recognised text uses the normal send path	Server-side speech-to-text	No audio stored or uploaded by TaskFlow; same permission checks (BR-TFA-14)
D-TFA-10	Audit stores outcomes only (user, kind, outcome, time)	Store full prompts and answers	No message bodies, passwords or keys in the audit (BR-TFA-13)

14. Risks, Assumptions & Open Issues

Type	Description	Mitigation / owner
Risk	Wrong or invented answer (R-TFA-01)	Help only from topics; ranking in code; footer warning
Risk	Prompt injection through task, comment or mail text (R-TFA-02)	Data marking, redaction, confirm and permission re-check
Risk	Company data sent to a cloud provider (R-TFA-03)	Cloud notice; local Ollama; secret screen
Risk	Conversations kept indefinitely (R-TFA-04)	⚠️ To be confirmed: retention period or "delete my history" – Jojo George
Open issue	⚠️ To be confirmed: role change checks only `roles.manage`, while Employee Management also needs `employees.role.assign` (and `roles.administrator.assign` for Administrator)	Jojo George
Open issue	⚠️ To be confirmed: Confirm / Edit / Cancel are not hidden for users without `assistant.actions.confirm`	Jojo George
Open issue	⚠️ To be confirmed: role-restricted help topics and "my manager" match by role name, so custom roles are treated as others	Jojo George
Open issue	⚠️ To be confirmed: password guidance text still mentions the current password (R-TFA-06)	Jojo George
Assumption	An administrator configures at least one provider; users understand AI answers can be wrong	Jojo George

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15. Traceability (requirements → components)

BRD/FRD item	Module / component	Section
FR-TFA-01 – FR-TFA-02	TFA Panel › `TfaPanelComponent` mounted by `ShellComponent` with `assistant.chat`; `AssistantController` policies	§5, §9
FR-TFA-03 – FR-TFA-05	TFA Panel › conversation load (last 100), suggestions, `TfaComposerComponent`	§5
FR-TFA-06	Own-scope repositories; 404 for foreign ids	§9
FR-TFA-07 – FR-TFA-08	Conversation engine › length check, `SecretScreen`	§8.1, D-TFA-6
FR-TFA-09	Task answers › `GetMyTasksTool` via `ITaskService`	§5.1
FR-TFA-10 – FR-TFA-11	Work order › `PrioritiseTasksTool`, `TaskPrioritiser`; ranked list in `TfaMessageComponent`	D-TFA-4
FR-TFA-12 – FR-TFA-14	Model selector (`TfaHeaderComponent`), providers action, `ChatModelGateway`	§7, D-TFA-7
FR-TFA-15	`TfaPanelComponent` cloud notice (`tfa.cloudAck`)	§9
FR-TFA-16	`ProviderFailure` classification, `ProviderFailureMessages`, failure card	§8.3
FR-TFA-17 – FR-TFA-19	Conversation engine › new chat, history window, tool-round limit	§6.3, §12
FR-TFA-20	Tool output marked as data and redacted	§9, D-TFA-1
FR-TFA-21	`AssistantService` (TS) resets on user change	§5
FR-TFA-22	`AssistantPrompt` with application name and timezone	§5
FR-TFA-23 – FR-TFA-24	Fixed refusal text (`AssistantTexts`); panel footer	§9
FR-TFA-25 – FR-TFA-28	How-to help › `SearchHelpTool`, `HelpKnowledgeBase` (19 topics, role-aware, screen links)	§5, D-TFA-5
FR-TFA-29 – FR-TFA-33	Action Cards › action tools, `AssistantActionService` (prepare, re-check, expiry, single claim, cancel), `TfaActionCardComponent`	§8.2, D-TFA-2, D-TFA-3
FR-TFA-34 – FR-TFA-38	Mail draft › `DraftMailTool`, edit action, `SendMailRequestValidator`, `IMailService.SendAsync`	§5.1
FR-TFA-39	Profile card › `ProposeProfileUpdateTool`, `IProfileService.UpdateAsync`	§5.1
FR-TFA-40 – FR-TFA-41	Role card › `ProposeRoleChangeTool` (`roles.manage`), `IEmployeeService.UpdateAsync`, sessions ended	§5.1, §14
FR-TFA-42 – FR-TFA-44	Lookup and guidance › `FindUserTool`, `ExplainPasswordChangeTool`, `ContactAdminGuidanceTool`	§5
FR-TFA-45 – FR-TFA-46	`AssistantActionService` failure status; card error messages	§7
FR-TFA-47 – FR-TFA-52	Voice controls › `WakePhraseService`, `SpeakService`, `TfaComposerComponent` mic	§8.4, D-TFA-9
FR-TFA-53	`AssistantRateLimiter`	§9, D-TFA-8

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BRD/FRD item	Module / component	Section
FR-TFA-54 – FR-TFA-56	`AssistantAuditLog` and audit repository	§9, D-TFA-10
NFR-TFA-01 – NFR-TFA-08	Non-functional design	§12
BR-TFA-01 – BR-TFA-14	Covered through the FR rows above (see FRD §11)	§15

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